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Identifying who are unlikely to benefit from total knee arthroplasty using machine learning models

Xiaodi Liu[1*], Yingnan Liu[1*], Mong Li Lee[1], Wynne Hsu[1], Ming Han Lincoln Liow[2]
(* equal contribution)
[1]National University of Singapore, [2]Singapore General Hospital

Introduction

- Knee osteoarthritis (OA) is the most common degenerative joint disease which leads to **significant disability** in the elderly, with a global **prevalence** of around **23%** in individuals aged over 40 years.
- Total Knee Arthroplasty (TKA) is a standard treatment for advanced OA.**15-20%** of patients report **dissatisfaction post-surgery**.
- The outcome of TKA is commonly **assessed** objectively with **patient-reported outcome measures** (PROMs) to measure baseline **function** preoperatively and patient improvement postoperatively.
- **Minimal clinically important difference** (MCID) thresholds of PROMs can be used to determine if a patient achieves perceivable benefits.
- Aim to develop and compare the performance of ML models using **radiographs** and **clinical data** to predict if MCID thresholds are met after TKA at 6-month and 2-year follow-ups.

Data

5,720 knee OA patients

- 7,224 anterior-posterior knee radiographs and clinical records

PROMs to study

- Knee Society Knee and Function Scores (KSS)
- Oxford Knee Score (OKS)
- Short Form-36 Health Survey Physical and Mental Component Score (SF-36 PCS, SF-36 MCS)

Preprocessing

- Radiograph: BoneFinder@ tool to extract **region** of interest of the **knee joint**, contrast-limited adaptive histogram equalization to **enhance image contrast**
- Clinical data: **normalization** for continuous variables, **one-hot encoding** for categorical variables
- Split data at **patient level** to training and testing sets (80:20)
- **Balanced sampling** to handle class imbalance between reach/not reach MCID

Methodology

A. Image-only model

- **ConvNext-Tiny** pretrained on ImageNet as CNN backbone
- Best among ResNet101, ResNeXt, ConvNeXt, Vision Transformer

B. Clinical-data only model

- **XGB** as learning algorithm
- Best among Random Forest, Support Vector Machine, decision tree MLP

C. Multimodal model

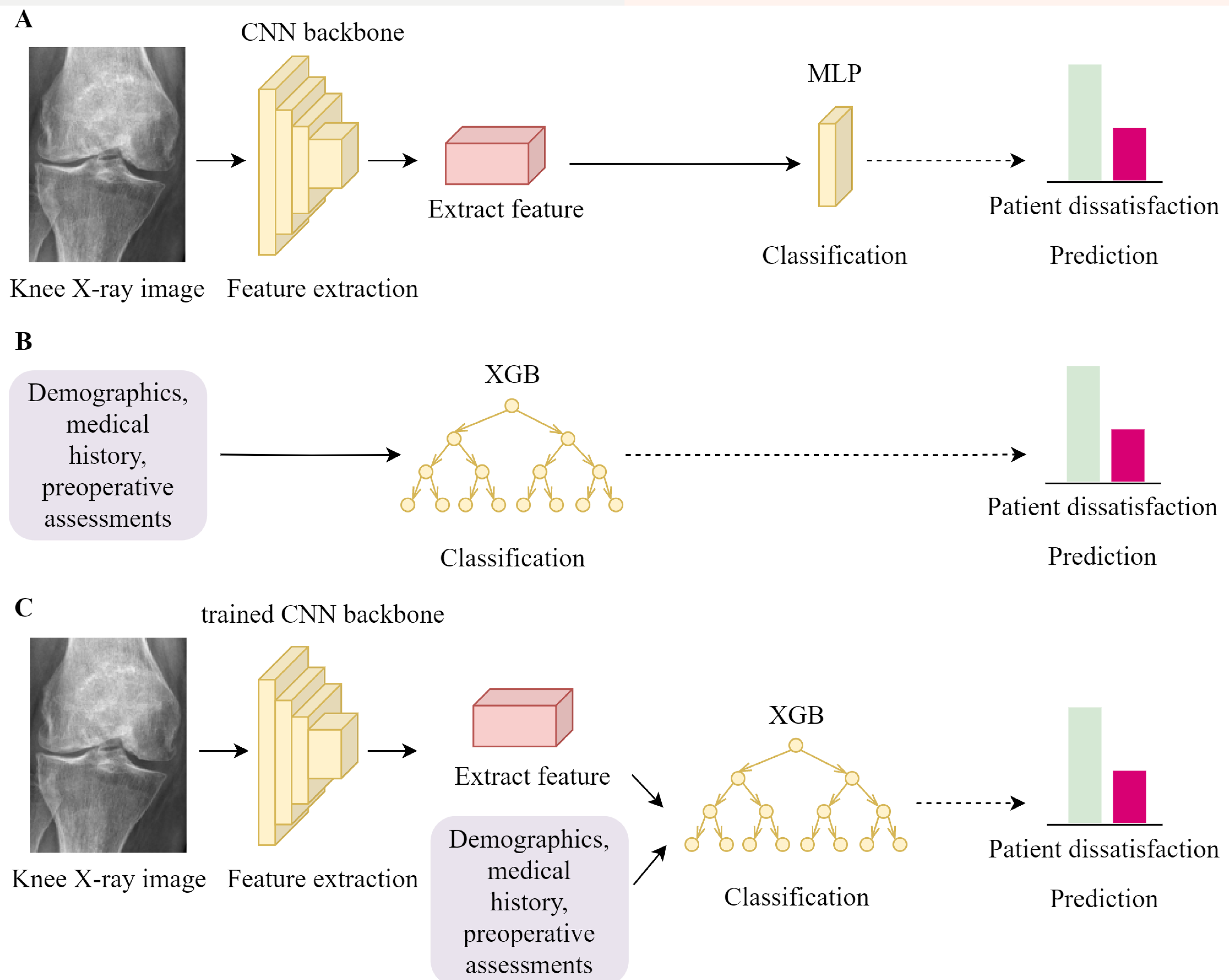
- Jointly train a CNN with clinical data to extract **complementary** image features
- XGB to learn prediction using learned image feature and clinical data

Evaluation

- **Cross validation** within the training set to tune hyperparameters. Test in the **internal testing set**.
- AUC, F1 score, Precision, Recall

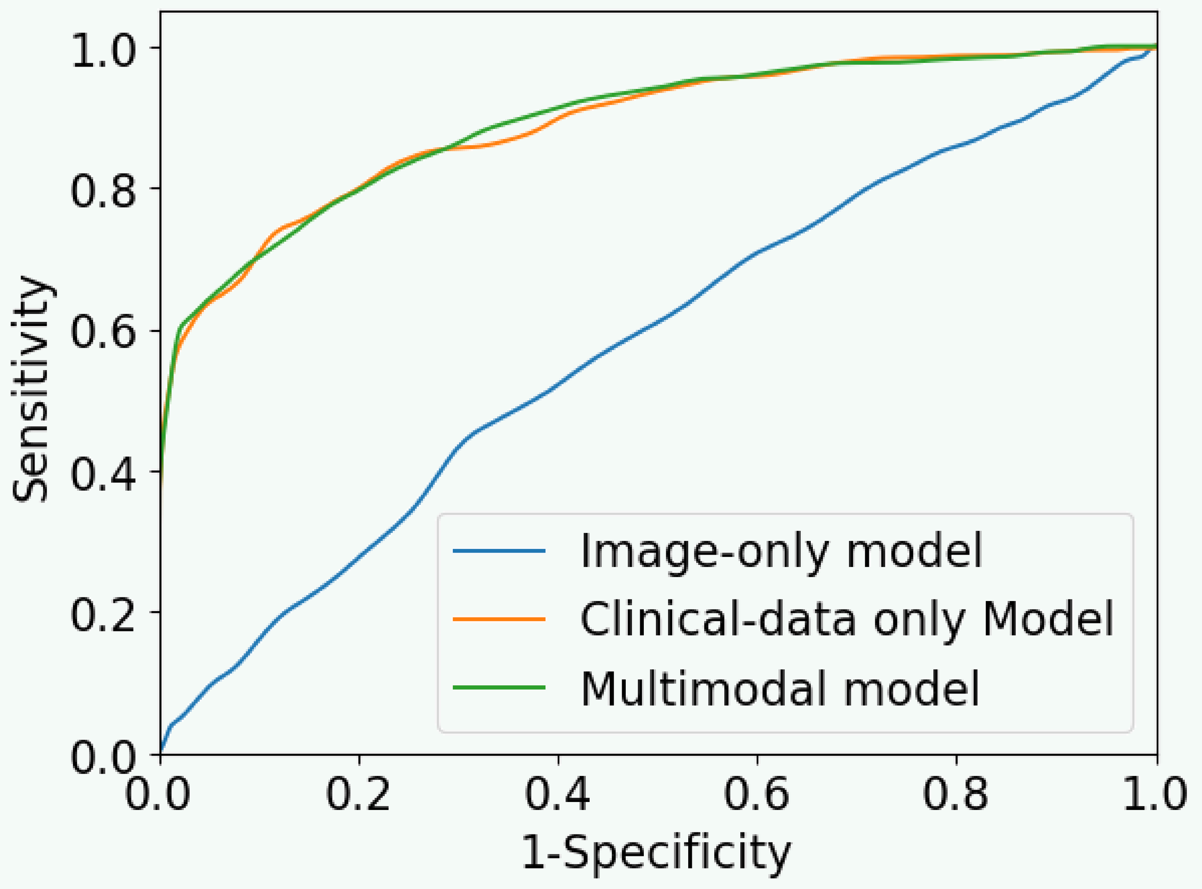
Statistical analysis

- N-out-of-n bootstrap with replacement to estimate 95% **confidence interval**
- DeLong's method to get **P values**



Result

- **Clinical-data only** and **multimodal** models achieved superior performance, compared to image-only model
- Multimodal model does **not statistically significantly outperform** clinical-data only model

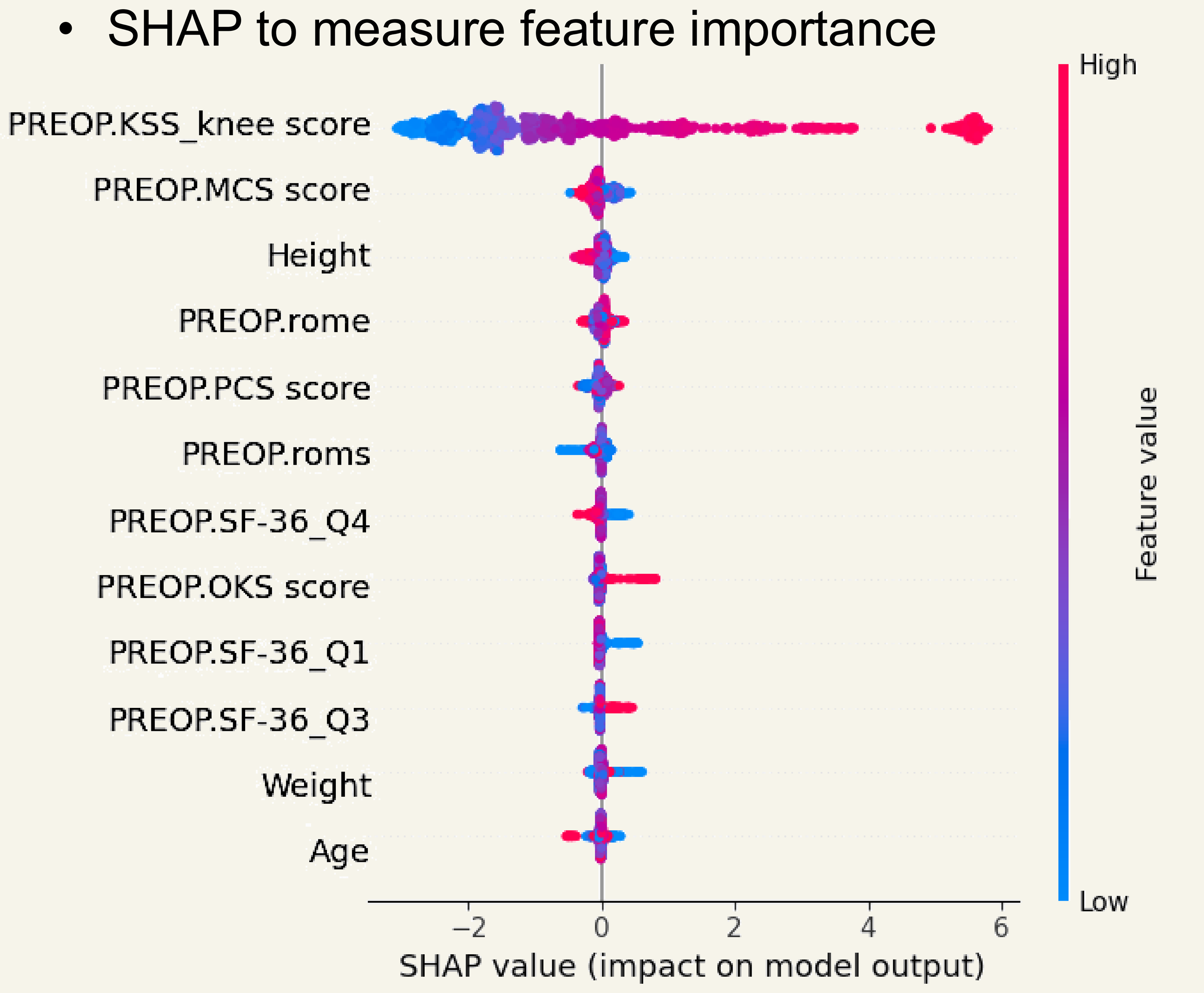


ROC curves for predicting not achieve KSS MCID at 2-year follow-up

Discussion

Preoperative scores are the most important input variable for predicting postoperative dissatisfaction

- SHAP to measure feature importance



SHAP explanation of clinical-data only model for predicting not achieve KSS MCID at 2-year follow-up

Pathological changes can be identified from images, but these features (presence/severity) cannot predict patient dissatisfaction

- GradCAM to visualize model's focus
- Similar severity of radiographic features lead to different satisfaction outcome

Not achieve KSS MCID | Achieve KSS MCID | Achieve KSS MCID

Paper

Code